Adaptive Sensor Fusion for Human Activity Recognition

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Abstract

This project investigates an adaptive sensor-fusion AI system for Human Activity Recognition (HAR) using smartphone motion sensor measurements from the MotionSense dataset. The project aims to recognize physical activities using sensor information while investigating how recognition performance changes when sensor measurements become noisy, unreliable, or unavailable.

Phase 1 defines the problem, identifies the research gap to be investigated, and proposes a machine-learning-based approach. Phase 2 focuses on data acquisition, cleaning, outlier management, feature preprocessing, and exploratory data analysis. The resulting pipeline prepares sensor measurements for subsequent machine learning experiments. Future work will evaluate whether adaptive sensor fusion improves robustness under degraded sensor conditions compared with a non-adaptive baseline.

Contents

1 Introduction **4**

2 Phase 1: Problem Definition **4**

2.1 Project Overview 4
2.2 Domain 4
2.3 Problem Statement 5
2.4 Target Audience 5
2.5 AI Justification 6
2.6 Research Gap 6
2.7 Project Objective 6
2.8 Proposed Approach 7
2.9 Scope and Limitations 7

3 Dataset Description **8**

3.1 Dataset Sources 8
3.2 Sensor Features 8

4 Phase 2: Data Acquisition and Preprocessing **8**

4.1 Data Acquisition 8
4.2 Data Cleaning 9
4.3 Outlier Management 9
4.4 Feature Preprocessing 9
4.5 Exploratory Data Analysis 9
4.5.1 Activity Distribution 9

4.5.2 Sensor Correlation 10
4.5.3 Sensor Feature Distributions 11

5 Results and Discussion **11**

6 Limitations and Future Work **12**

7 Conclusion **12**

8 References **12**

1 Introduction

Human Activity Recognition (HAR) involves identifying physical activities using sensor measurements. Smartphone motion sensors, including accelerometers and gyroscopes, capture movement information that can help distinguish activities such as walking, jogging, sitting, standing, and moving upstairs or downstairs.

Combining information from multiple sensors can improve activity recognition because different measurements describe different aspects of movement. However, sensor readings may become noisy, unreliable, or unavailable because of device movement, poor placement, temporary data loss, or hardware problems.

This project investigates an adaptive sensor-fusion system designed to adjust its use of sensor information under changing sensor conditions. The work is organized into phases, beginning with problem definition and continuing with data acquisition and preprocessing before model development and evaluation.

2 Phase 1: Problem Definition

2.1 Project Overview

This project proposes a machine-learning-based system for Human Activity Recognition using smartphone motion sensor data. The system aims to classify six activities: downstairs, upstairs, walking, jogging, standing, and sitting.

The proposed approach combines information from motion sensor features and investigates how the system can respond when some sensor measurements become noisy or unavailable. Under normal conditions, the system can use the available sensor information. Under degraded conditions, an adaptive strategy may adjust the contribution of different sensor groups or rely more heavily on the remaining reliable information.

The project will compare the proposed adaptive approach with a non-adaptive baseline to determine whether adaptation improves recognition performance under the tested conditions.

2.2 Domain

The project falls within the following domains:

- Artificial Intelligence and Machine Learning.
- Human Activity Recognition.
- Sensor Fusion and Robust Machine Learning.
- Wearable Computing and Smartphone-Based Activity Tracking.

These domains involve using sensor measurements and computational models to identify physical activities and investigate the reliability of predictions under changing data conditions.

2.3 Problem Statement

Human Activity Recognition systems use sensor measurements to identify a person's physical activities. Accelerometers measure linear acceleration, while gyroscopes measure rotational movement. Combining information from these sensors can help a model distinguish between different activities.

In practical situations, sensor readings may be affected by device movement, poor placement, measurement noise, temporary data loss, or hardware problems. If a system depends heavily on all available sensor measurements, degraded readings from one sensor may reduce recognition accuracy or lead to incorrect predictions.

For example, noisy rotational measurements may make activity classification more difficult. Similarly, if measurements from one sensor group become unavailable, a model that relies on those measurements may perform poorly.

The problem addressed by this project is how to maintain reliable human activity recognition when sensor measurements become noisy, unreliable, or unavailable. The proposed system will investigate strategies for adjusting the use of available sensor information under such conditions and evaluate their performance against a non-adaptive baseline.

2.4 Target Audience

The intended users and beneficiaries of this system include:

- **Developers of wearable and smartphone applications:** Developers who need activity recognition to remain reliable when sensor data quality changes.
- **Researchers:** Researchers studying human activity recognition, sensor fusion, and robust machine learning who want to evaluate methods for handling noisy or missing sensor data.
- **Fitness and wellness application teams:** Teams using activity classification to summarize everyday movement. Real-world deployment would require additional testing and validation.
- **Students and project teams:** Students exploring adaptive AI systems and sensor-based machine learning applications.

2.5 AI Justification

Machine learning is suitable for this problem because motion sensor measurements contain patterns that vary across activities, individuals, and sensor conditions. A machine-learning model can learn relationships between labelled sensor measurements and activities rather than relying entirely on manually defined rules.

A traditional rule-based system might use fixed acceleration thresholds to identify an activity. However, the same thresholds may not work consistently across different movement patterns or noisy measurements. Maintaining rules for every possible activity and sensor condition can also become difficult.

A classifier such as Random Forest can learn patterns from sensor features and their corresponding activity labels. Sensor fusion allows the system to use information from multiple sensor groups, while an adaptive strategy can change how predictions are formed when some measurements are degraded or missing.

For example, the system could use information from both accelerometer and gyroscope measurements when they are available and reliable. If one group becomes unavailable, a suitable alternative strategy could rely on the remaining measurements.

The effectiveness of this approach cannot be assumed in advance. It must be established experimentally by comparing the baseline and adaptive approaches under normal and degraded sensor conditions.

2.6 Research Gap

Human Activity Recognition performance can be affected when sensor measurements become noisy, unreliable, or unavailable. This project investigates whether an adaptive sensor-fusion approach can reduce the impact of such conditions compared with a non-adaptive baseline.

The research question is:

> Can an adaptive sensor-fusion approach maintain better human activity recognition performance than a non-adaptive baseline when selected sensor measurements become noisy or unavailable?

The project will investigate this question using controlled sensor degradation experiments. The research gap is treated as a question to evaluate, rather than an assumption that existing systems cannot handle sensor failures. A literature review would be required to establish how the proposed approach differs from existing methods.

2.7 Project Objective

The main objective is to develop and evaluate a machine-learning-based human activity recognition system that combines motion sensor information and adapts to noisy or miss-

ing measurements, with the goal of improving robustness compared with a non-adaptive baseline.

The specific objectives are:

- Acquire and prepare motion sensor data for activity recognition.
- Explore the distribution and characteristics of the sensor measurements.
- Develop a baseline human activity recognition model.
- Simulate noisy and unavailable sensor measurements.
- Investigate an adaptive sensor-fusion strategy.
- Compare performance under normal and degraded sensor conditions.
- Evaluate the approaches using accuracy, precision, recall, and F1-score.

2.8 Proposed Approach

The proposed system will use motion sensor features to classify human activities. A baseline classifier will first be developed using the available sensor features.

Sensor degradation will then be simulated by introducing noise or making selected sensor measurements unavailable. An adaptive fusion strategy will be investigated to adjust the contribution of different sensor groups or select an appropriate prediction strategy according to the available information.

The baseline and adaptive approaches will be compared under normal and degraded conditions using consistent evaluation metrics. The final model and adaptation strategy will depend on experimental results.

2.9 Scope and Limitations

The initial project focuses on the six activity classes available in the MotionSense dataset. The current phase covers data acquisition, cleaning, outlier management, feature preprocessing, and exploratory data analysis.

Future experiments will investigate simulated noisy and missing sensor measurements. Therefore, results will apply to the dataset, models, and degradation conditions tested. Simulated sensor failure does not necessarily reproduce every real hardware failure or environmental condition.

Further testing with different users, devices, and real sensor failures would be necessary before making broader claims about real-world reliability.

3 Dataset Description

The project uses the publicly available MotionSense dataset, which contains smartphone motion measurements associated with six activities: downstairs, upstairs, walking, jogging, standing, and sitting.

The dataset includes 12 sensor features covering device attitude, gravity, rotation rate, and user acceleration. Participant information is also integrated into the dataset.

3.1 Dataset Sources

- **Sensor dataset:** https://github.com/mmalekzadeh/motion-sense
- **Participant information:** https://raw.githubusercontent.com/mmalekzadeh/motion-sense/master/data/data_subjects_info.csv

3.2 Sensor Features

The 12 sensor features used in the current preprocessing pipeline are:

- **Attitude:** roll, pitch, and yaw.
- **Gravity:** x, y, and z components.
- **Rotation rate:** x, y, and z components.
- **User acceleration:** x, y, and z components.

These features provide information about device orientation, gravity, angular movement, and linear acceleration. All 12 features are retained and standardized during preprocessing so that their usefulness can be evaluated during later experiments.

4 Phase 2: Data Acquisition and Preprocessing

4.1 Data Acquisition

The pipeline downloads the MotionSense sensor dataset and participant metadata from the public repository. The sensor CSV files are combined into a structured dataset containing activity labels, participant identifiers, trial numbers, and source-file information.

The acquisition process also validates the downloaded files and generates a collection report.

4.2 Data Cleaning

Sensor measurements are converted to numeric values, and infinite and missing values are handled. Duplicate samples are removed, and participant metadata is merged with the sensor records.

The cleaning process generates a summary containing information about the processed records and data quality.

4.3 Outlier Management

The Interquartile Range (IQR) method is used to identify extreme sensor measurements. For each sensor feature, the first quartile, third quartile, and interquartile range are calculated.

Values outside the lower and upper IQR bounds are capped to reduce the influence of extreme measurements. The outlier analysis summary records the number of detected and capped values for each feature.

4.4 Feature Preprocessing

Activity labels are encoded into numeric values using LabelEncoder. All 12 sensor features are standardized using StandardScaler.

The standardized features are added to the dataset while the original sensor columns are retained. The fitted scaler and activity label mapping are saved for future use.

4.5 Exploratory Data Analysis

Exploratory Data Analysis (EDA) is performed to examine the activity distribution, sensor feature distributions, and relationships between sensor measurements.

4.5.1 Activity Distribution

Figure 1 presents the distribution of samples across activity classes. This visualization helps identify differences in the number of available observations for each activity.

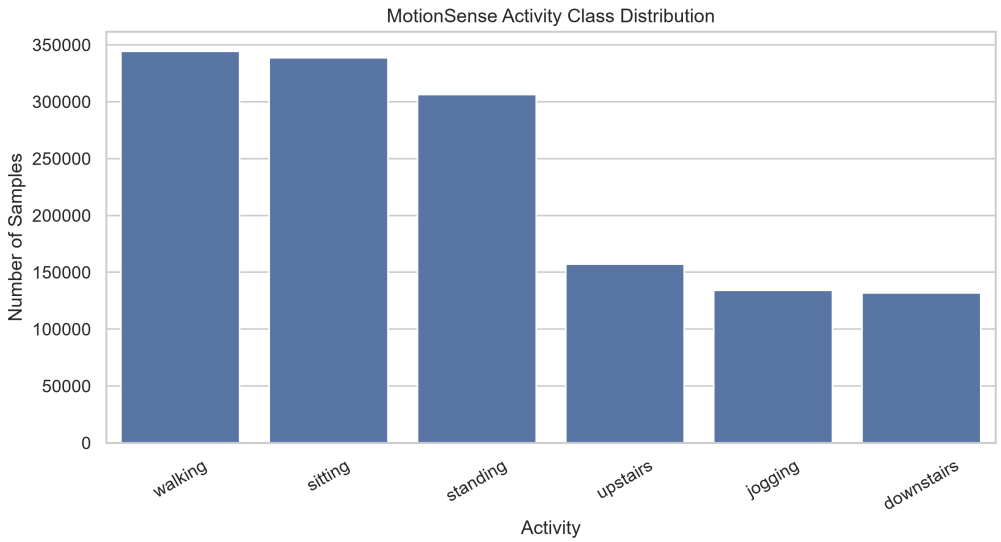


Figure 1: Distribution of samples across activity classes.

4.5.2 Sensor Correlation

Figure 2 shows the correlation between sensor features. The heatmap helps identify relationships between measurements that may be useful when selecting features and designing the sensor-fusion approach.

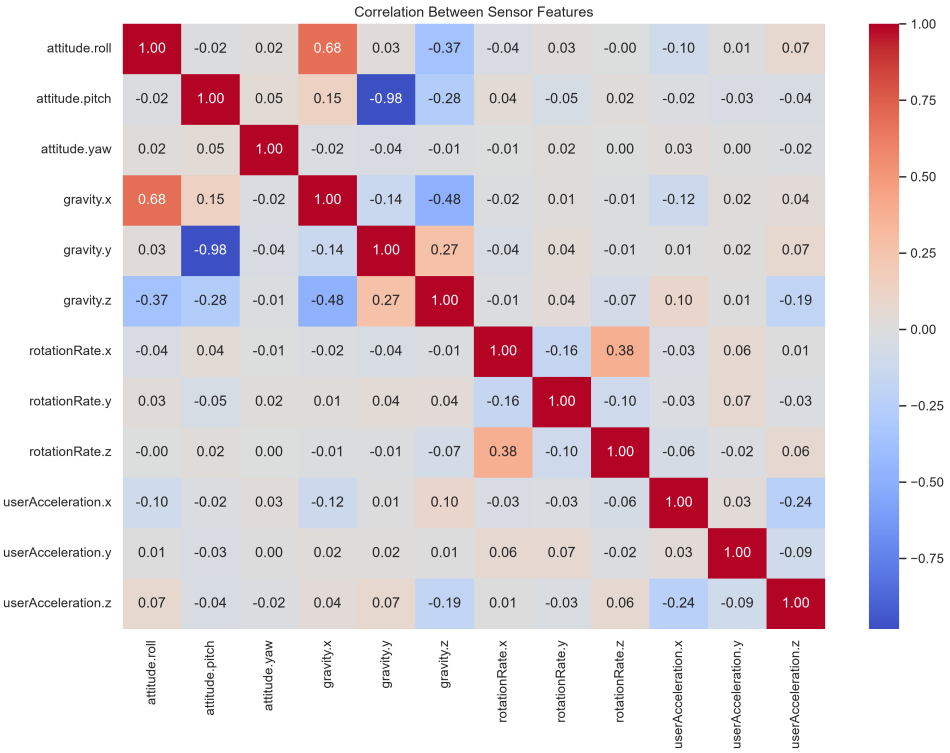


Figure 2: Correlation heatmap of sensor features.

4.5.3 Sensor Feature Distributions

Figure 3 presents the distributions of the sensor features. These distributions help examine the spread and patterns of the measurements after data cleaning and outlier management.

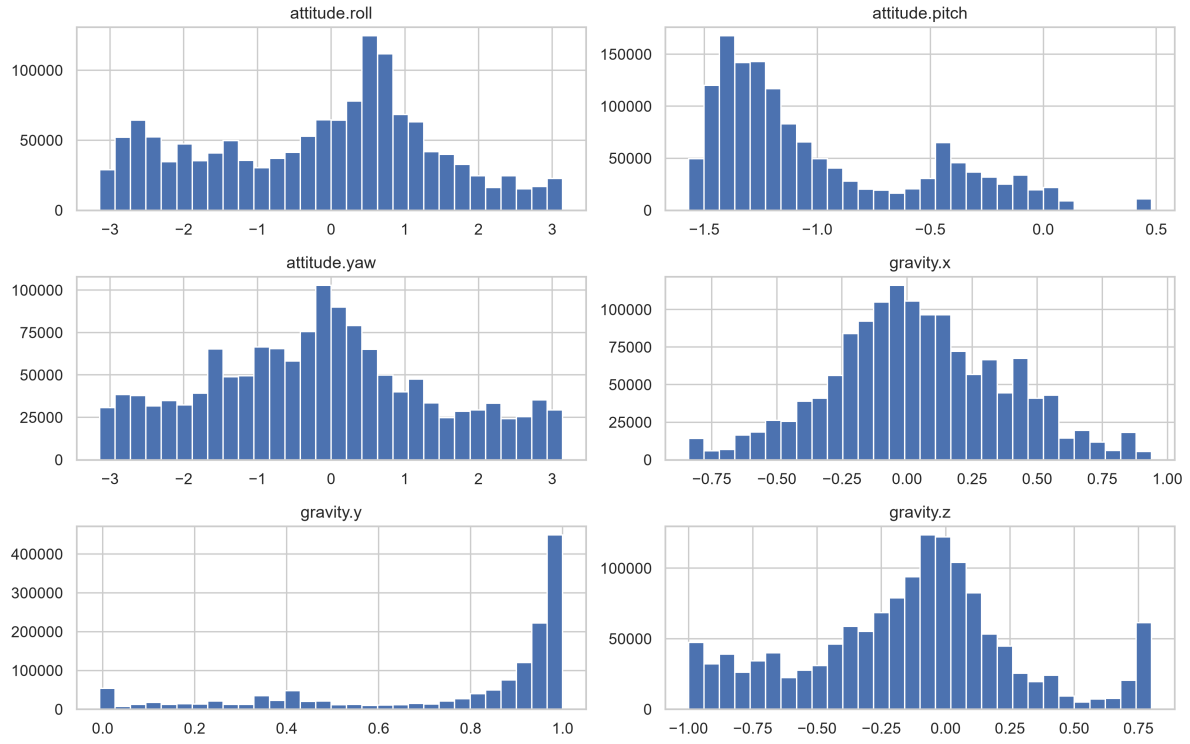


Figure 3: Distributions of the sensor features.

5 Results and Discussion

The Phase 2 pipeline completed data acquisition, cleaning, outlier management, feature preprocessing, and exploratory data analysis.

The acquisition report recorded 364 sensor CSV files and participant information for 24 participants. The initial cleaning stage processed approximately 1.4 million sensor records. The pipeline also generated outlier summaries, an activity label mapping, a fitted scaler, and EDA charts.

The activity distribution visualization can be used to examine differences in sample counts across activities. The correlation heatmap and feature distribution plots provide an initial understanding of the sensor measurements.

Quantitative conclusions about classification performance cannot yet be drawn because the baseline model and adaptive sensor-fusion experiments have not been evaluated in this phase. Those results will be added after the experiments have been implemented

and run.

6 Limitations and Future Work

The current work focuses on preparing and exploring the dataset. It does not yet establish whether adaptive sensor fusion improves activity recognition under sensor failure.

Future work includes:

- Reviewing existing research on adaptive sensor fusion and robust human activity recognition.
- Developing a baseline activity classification model.
- Simulating noise and missing sensor measurements.
- Implementing an adaptive sensor-fusion approach.
- Comparing baseline and adaptive models under normal and degraded sensor conditions.
- Evaluating performance using accuracy, precision, recall, and F1-score.
- Using an appropriate evaluation protocol to assess generalization to unseen data.

7 Conclusion

Phase 1 defined the problem of human activity recognition under unreliable sensor conditions, identified the research question, and proposed a machine-learning-based adaptive sensor-fusion approach.

Phase 2 established a reproducible pipeline for acquiring, cleaning, preprocessing, and exploring the MotionSense dataset. The pipeline prepares 12 sensor features for future machine learning experiments and generates visualizations that support initial data analysis.

These stages provide the foundation for developing and evaluating the proposed system. Further experiments are required to determine whether adaptive sensor fusion improves recognition performance when sensor measurements are degraded or unavailable.

8 References

1. Malekzadeh, M. et al. MotionSense dataset. Available at: https://github.com/mmalekzadeh/motion-sense

2. MotionSense participant information. Available at: https://raw.githubusercontent.com/mmalekzadeh/motion-sense/master/data/data_subjects_info.csv